

Lab 4 — Vibe Coding Chat History

62_scramble — Word Scramble Repair Lab

Vibe Coding Chat History — Lab 4

Task 1 — Fix validation bug

****Prompt****

Inspect the provided Pygame Word Scramble code. There is a bug in the guess validation: the player's answer is being compared against the scrambled word instead of the original secret word. Identify the cause and fix only this bug. Do not add new features or modify unrelated code.

****Result****

Changed:

`is\_correct = (guess == self.scrambled\_word)`

to:

`is\_correct = (guess == self.secret\_word)`

Task 2 — Hint

****Prompt****

Add a HINT button to the existing Word Scramble game. When clicked, reveal the first unrevealed letter of `secret\_word` in its correct position and deduct one point from the score, without changing the current round. Preserve the existing guess-validation behavior. Do not implement the timer or letter tiles yet.

****Result****

Added `HINT` button, revealed-letter state, first-unrevealed-letter logic, and a one-point penalty.

Task 3 — Countdown timer

****Prompt****

Add a 20-second countdown timer to each round of the existing Word Scramble game. Display a visible timer bar and remaining seconds. If the timer reaches zero before a correct answer is submitted, reveal the correct word, show `TIME'S UP!`, and automatically advance to the next round after a short message. Preserve the hint and guess-validation features. Do not implement letter tiles yet.

****Result****

Added a 20-second round timer, progress bar, timeout message, and automatic transition.

Task 4 — Letter tiles

****Prompt****

Add graphical letter tiles for the scrambled word. Each letter should be rendered in its own square tile. Let

the player click one tile and then another tile to swap their positions. Keep the tile order synchronized with the answer input so the player can submit the rearranged letters. Preserve the bug fix, hint system, and countdown timer.

****Result****

Added clickable tiles, tile swapping, selected-tile highlighting, and synchronization with the answer box.

Final review

The completed implementation contains all four requested tasks and keeps the original Pygame structure:

- `main.py`
- `game/game\_engine.py`
- `game/text\_box.py`